

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

593000613 - Programming Of Wireless Devices

DEGREE PROGRAMME

59AJ - Master Universitario En Comunicaciones Inalámbricas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	593000613 - Programming Of Wireless Devices
No of credits	4.5 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	59AJ - Master Universitario en Comunicaciones Inalámbricas
Centre	59 - E.T.S. De Ingeniería Y Sist. De Telecom.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Alberto Martin Perez	A7011	a.martinp@upm.es	Sin horario.
Carlos Cortes Sanchez (Subject coordinator)	D8418	carlos.cs@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- XML and/or JSON understanding
- Java programming skills

4. Skills and learning outcomes *

4.1. Skills to be learned

CB6 - Poseer y comprender conocimientos que aporten una base u oportunidad de ser originales en el desarrollo y/o aplicación de ideas, a menudo en un contexto de investigación/ Students have demonstrated knowledge and understanding providing the groundwork or opportunity for innovation in developing and/or applying ideas, often within a research context

CB7 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio/Students are capable of applying their knowledge, understanding, and problem-solving abilities in new or unfamiliar environments within broader (or multidisciplinary) contexts related to their field of study

CB8 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios/Students are capable of integrating knowledge and making complex decisions, which, although based on incomplete or limited information, require reflection on social and ethical responsibilities linked to the application of their knowledge and opinions

CGI02 - Comprender el procedimiento, valor y límites del método científico, siendo capaz de identificar, localizar y obtener datos requeridos en un trabajo de investigación, de diseñar y guiar investigaciones analíticas, de modelado y experimentales, así como de evaluar datos de una manera crítica y extraer conclusiones. / Understand the procedure, value, and limits of the scientific method, being able to identify, locate and obtain data required in a

research work, to design and guide analytical, modeling, and experimental investigations, as well as to critically evaluate data and extract conclusions.

CGI03 - Valorar la importancia de las fuentes documentales, manejarlas y buscar la información para el desarrollo de cualquier trabajo de investigación. / Assess the importance of documentary sources, manage them and search for information for the development of any research work.

CGI04 - Leer y comprender publicaciones dentro de su ámbito de estudio/investigación, así como su catalogación y valor científico. / Read and understand publications within their field of study / research, as well as their cataloging and scientific value.

UPM1 - Uso de la lengua inglesa / Use of the English language

UPM4 - Organización y planificación / Planning and organization

4.2. Learning outcomes

RA32 - Programar el comportamiento de los sensores de un equipo de usuario y extraer la información recogida por ellos/ Program the behavior of the sensors of a user equipment and extract the information collected by them

RA33 - Identificar los elementos del código fuente de un programa y modificar lo necesario para conseguir un objetivo determinado/ Identify the elements of the source code of a program and modify what is necessary to achieve a specific objective

RA34 - Aplicar el uso de interfaces de usuario y de persona-máquina para mejorar la usabilidad de un producto/ Apply the use of user and human-machine interfaces to improve the usability of a product.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

No hay descripción de la asignatura.

5.2. Syllabus

1. Introduction and basic Android User Interfaces

1.1. Introduction and IDE

1.2. Basic UI Events

1.3. Event Processing

2. Sensor Data Access

2.1. Common Mobile Sensors

2.2. Movement and Location Sensors

2.3. Sensor Acquisition System

3. Sensor Communications

3.1. Wired Protocols

3.2. Wireless Protocols

3.3. Development Kits

4. Communications and Data Processing

4.1. Network Interfaces

4.2. Messaging Protocols for Sensor Networks

4.3. Use Case: Message Queue Telemetry Transport

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction. Android IDE. Basic Android UI. Android sensor framework. Duration: 02:00	Android Framework, Basic APPs, Sensor framework: practice Duration: 02:00		
2	Sensor Data Access Duration: 02:00	Read embedded thermal sensor Duration: 02:00		
3	Sensor Communications: Wired Protocols Duration: 02:00	Read an external sensor using a wired communication protocol Duration: 02:00		
4	Sensor Communications: Wireless Protocols Duration: 02:00	Read an external sensor using a wireless protocol Duration: 02:00		
5	Sensor Networks Duration: 02:00	Energy consumption measurements in devices Duration: 02:00		
6		Practical works to exercise the latest learnt contents Duration: 04:00		
7	Network Interfaces and Messaging Duration: 02:00	Data visualization and MQTT guided exercises. Duration: 02:00		
8				First theoretical exam Progressive assessment and Global Examination Presential Duration: 02:00
9		Project work in groups Duration: 02:00		
10		Project work in groups Duration: 02:00		
11		Project work in groups Duration: 02:00		

12		Project work in groups Duration: 02:00		
13		Project works in groups Duration: 02:00		
14		Project works in groups Duration: 02:00		
15				Project defense Progressive assessment and Global Examination Presential Duration: 02:00
16				
17				

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
8	First theoretical exam		Face-to-face	02:00	30%	5 / 10	CB7 CGI02 UPM1 UPM4
15	Project defense		Face-to-face	02:00	70%	5 / 10	CB6 CB7 CB8 CGI02 CGI03 CGI04 UPM1 UPM4

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
8	First theoretical exam		Face-to-face	02:00	30%	5 / 10	CB7 CGI02 UPM1 UPM4
15	Project defense		Face-to-face	02:00	70%	5 / 10	CB6 CB7 CB8 CGI02 CGI03 CGI04 UPM1 UPM4

7.1.3. Referred (re-sit) examination

No se ha definido la evaluación extraordinaria.

7.2. Assessment criteria

The lectures given by the teachers will provide students with the basic knowledge on the design and implementation of mobile apps for IoT environments. Taking this knowledge as a starting point, students have to do both guided exercises and more self-guided elaborated projects individually and in groups. To perform these projects, students will receive assistance from the teachers, in class and in supervision sessions. The evaluation of this work will be done by assessing the code produced, a written report and an oral presentation done in front of the class, as well as an intermediate individual assessment.

At the beginning of the teaching period, each student can choose between two evaluation itineraries: Continuous evaluation itinerary: this is the option by default. In this itinerary, attendance to class is compulsory and the students will receive guidance during the development of their projects. To choose the option of the just final exam itinerary, it is necessary that the student requests it in written form, no later than one month after the beginning of the classes, and registers this application to the subject coordinator. In this case, it is understood that the student waives the right to receive guidance in class for their project, and attendance to class is not compulsory.

Regardless of the chosen itinerary, all students will have to do the practical work proposed by the teachers throughout the class period, produce the written report and participate in the oral presentation of their project. Besides, attendance to the oral presentations of the rest of the groups is compulsory for both itineraries.

The final grade for the course will be provided by the teachers. It will be obtained taking into account a) the exams done individually by the students, b) the project to be carried out in groups, including its code and the generated documentation and c) the oral presentation of the project. The weights of these items are specified in the tables above. If a student does not pass the subject in the ordinary period, they will have the opportunity of going through an additional evaluation process during the extraordinary period on July. The extraordinary assessment is only granted if the practical work described above has been completed, the project report has been produced, and participation has taken place in the oral presentation of yourself and the rest of the groups. The extraordinary exam will consist of two parts: firstly, a questionnaire selected among all topics in this course must be responded on-line

and, secondly, a lab practice with several exercises related to the practical activities carried out along the course must be duly done. Finally, a design proposal will be defended orally. In the extraordinary exam It will be necessary to pass each part before moving on to the next part.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Android developers website	Web resource	https://developer.android.com/
Android Studio: download and user guide	Web resource	https://developer.android.com/studio
Android Programming for Beginners	Bibliography	HORTON, J., 2021. Android Programming for Beginners - Third edition. Packt Publishing. Free online access for UPM students and staff.
Android Sensor Programming By Example	Bibliography	NAGPAL, V., 2016. Android sensor programming by example: take your Android applications to the next level of interactivity by exploring the wide variety of Android sensors. Packt Publishing. (Free online access for UPM students and staff)
Mobile sensors and context-aware computing	Bibliography	Gajjar, Manish J., author. 1st edition. Cambridge, Massachusetts : Elsevier 2017 Free online access for UPM students and staff

Beginning MicroPython with the Raspberry Pi Pico: Build Electronics and IoT Projects	Bibliography	Bell, Charles 1 Berkeley, CA: Apress 2022 Free online access for UPM students and staff
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